DESIGN NOTE

# AI Oversight: three-stage extraction pipeline

ai_oversight turns an AI-governance report (PDF or PPTX) into 22 canonical fields defined by PLAN.md — submission ID, model owner, risk rating, and so on — each with a value, a citation back to the source document, a status, and a confidence score. The pipeline runs in three stages, and the second and third each only do real work when the cheaper stage before them couldn't finish the job.

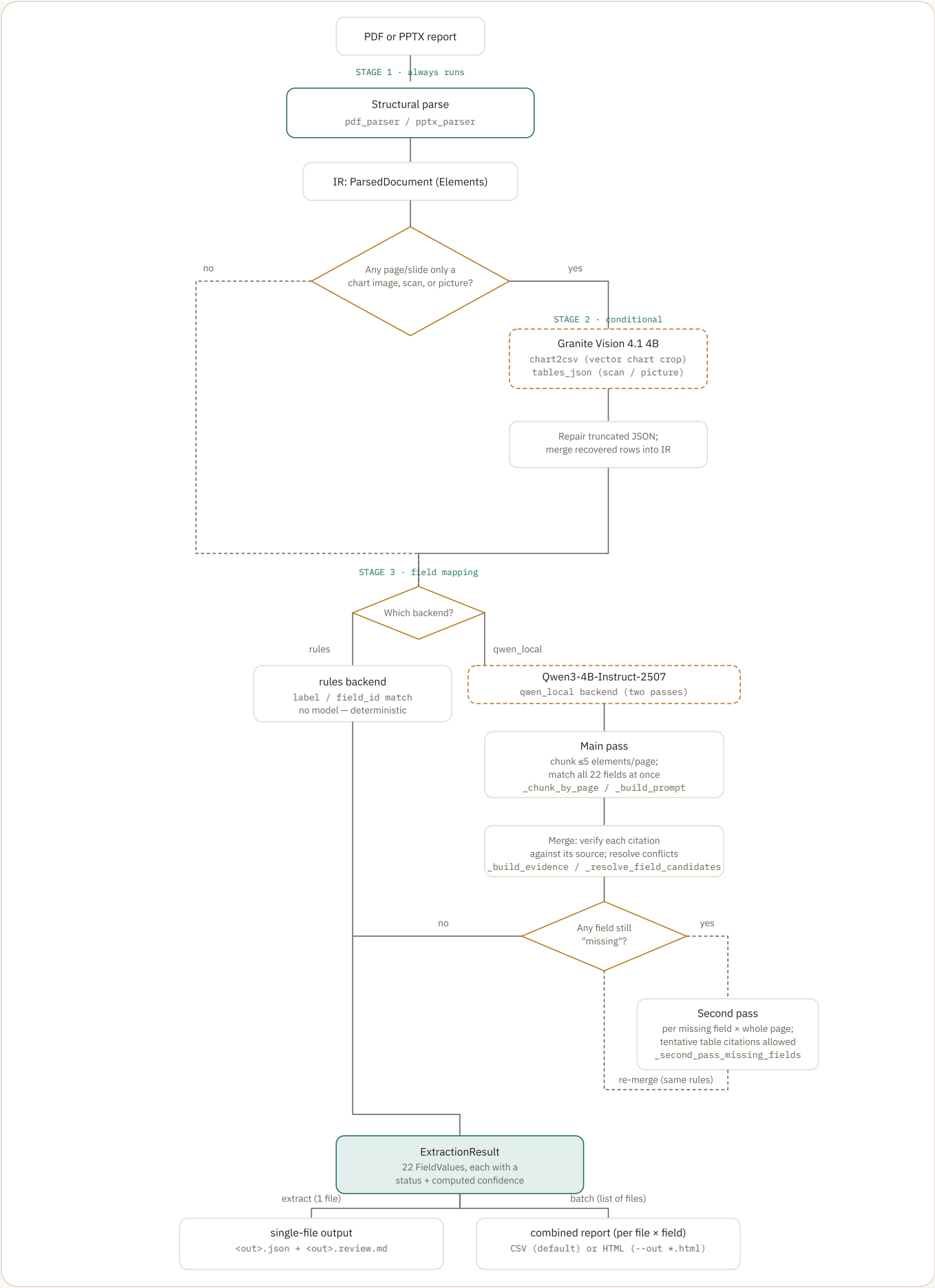
The organizing rule throughout is **never fabricate**: a value is only ever reported as confidently `extracted` when it can be traced to real text in the source document. Every other outcome — `needs_review`, `ambiguous`, `conflicting`, `missing` — exists so that uncertainty is visible to a human reviewer instead of silently guessed away.

Both models — Granite Vision 4.1 4B and Qwen3-4B-Instruct-2507 — are downloaded once from Hugging Face and run entirely on local hardware. There's no API call and no per-token cost to run `extract`, no matter how many times a document is processed or re-processed.

Hugging Face hosts many alternative models that could fill either role — larger, more capable ones included. Qwen3-4B-Instruct-2507 was chosen specifically for its small size: it keeps GPU and memory requirements low enough to run comfortably on ordinary local hardware (this project targets a 16GB Mac), rather than requiring a dedicated GPU server. Swapping in a different `AI_OVERSIGHT_TEXT_MODEL` is a config change, not a code change (see `Settings` below) — but every larger model trades away some of that low-resource portability.

## The pipeline, end to end

Stage 1 is always cheap and deterministic. Stage 2 (Granite Vision) only runs on the specific pages/slides Stage 1 flagged as ambiguous — a vector-drawn chart, a scanned page, or a picture shape — never on content Stage 1 already parsed cleanly. Stage 3 maps whatever elements exist (native + vision-recovered) onto the 22 canonical fields, using either a fast deterministic backend or a two-pass local LLM backend.


PDF or PPTX report
STAGE 1 · always runs
Structural parse
pdf_parser / pptx_parser
IR: ParsedDocument (Elements)
Any page/slide only a chart image, scan, or picture?
no
yes
STAGE 2 · conditional
Granite Vision 4.1 4B
chart2csv (vector chart crop)
tables_json (scan / picture)
Repair truncated JSON; merge recovered rows into IR
STAGE 3 · field mapping
Which backend?
rules
qwen_local
rules backend
label / field_id match
no model — deterministic
Qwen3-4B-Instruct-2507
qwen_local backend (two passes)
Main pass
chunk ≤5 elements/page;
match all 22 fields at once
_chunk_by_page / _build_prompt
Merge: verify each citation against its source; resolve conflicts
_build_evidence / _resolve_field_candidates
Any field still "missing"?
no
yes
Second pass
per missing field × whole page;
tentative table citations allowed
_second_pass_missing_fields
re-merge (same rules)
ExtractionResult
22 FieldValues, each with a status + computed confidence
extract (1 file)
batch (list of files)
single-file output
<out>.json + <out>.review.md
combined report (per file × field)
CSV (default) or HTML (--out *.html)

Dashed outlines/edges mark the two conditional branches: Stage 2 only runs for flagged pages, and the second qwen_local pass only runs for fields the main pass left `missing`.

## Stage 1 — Structural parse

`pdf_parser.parse()` (PyMuPDF) and `pptx_parser.parse()` (python-pptx) turn the raw file into the shared intermediate representation: a flat list of `Element`s, each one `text`, `key_value`, `table`, or `chart`. No model runs here — everything is regex, geometry, and shape structure.

Every stage exchanges data as Pydantic models — `Element`, `ParsedDocument`, `VisionObservation`, `ExtractionResult` — each one round-trips losslessly through `model_dump_json()` / `model_validate_json()`. That's deliberate: Stage 1's job is to capture everything the source document states, not to decide yet what's relevant to which of the 22 fields — that decision is Stage 3's alone. A saved `ParsedDocument` JSON carries the same information no matter which backend or model eventually reads it.

**PDF specifics**

Column-aware text blocks (the source reports are two-column); label→value pairing gated by on-page distance so a pairing is never guessed across unrelated text; a vector-drawn chart's gridline cluster is detected and its bounding region computed for Stage 2 to crop later.

**PPTX specifics**

Numbered-ID shape rows become `key_value` elements; non-ID multi-shape rows become `table` elements (ambiguous, multi-cell); a picture shape is never silently dropped — it's recorded as a warning that routes it to Stage 2.

This stage also decides, implicitly, whether Stage 2 has anything to do: a page that parsed cleanly carries no warning and is never sent to the vision model.

## Stage 2 — Granite Vision (conditional)

`granite_vision.run_vision_pass()` only fires for exactly three situations Stage 1 could flag but not resolve on its own:

| TRIGGER | TASK SENT TO GRANITE | WHY |
|---|---|---|
| A page has a vector-drawn chart (gridlines detected, not a native chart object) | `chart2csv` | PyMuPDF can see the strokes but not what they encode |
| A PDF page has no text layer and contains an image (fully scanned page) | `tables_json` | no native content exists to fall back on |
| A PPTX picture shape | `tables_json` | can't tell chart from table from a bare image, so the more general task is the default |

Granite's raw response is repaired if the model's EOS token cut it off mid-table ( `_repair_truncated_json` recovers every complete row before the cutoff), then merged back into the IR as ordinary `Element` s — a recovered chart becomes a `chart` element, a single clean instance row becomes a `key_value` , and a genuinely multi-instance table stays a cautioned `table` element. Whichever Stage 3 backend runs next never knows the difference between native and vision-recovered content.

Because this stage is slow (a real local vision model, also free of any API or token cost to run), its output can be cached once with the `vision` CLI command and replayed via `extract --vision-cache` — Stage 3 can then be iterated on repeatedly without re-running Granite.

The same applies one level up: Stage 1's own output can be saved with `parse --out` , and that JSON is exactly the input every Stage 3 backend consumes. A saved parse — with or without a saved vision cache merged into it — can be re-matched by `rules` , by `qwen_local` , or by a future LLM backend, swapping which model does the matching without ever re-parsing the document or re-running Granite.

## Stage 3 — Field mapping

Two interchangeable backends consume the same IR and produce the same `ExtractionResult` shape.

### rules backend — deterministic

Matches each `key_value` / `table` element's own numbered `field_id` or label text against the canonical 22, with a synonym table for common wording variants. No model, no network, effectively instant — the right choice when the source document's own numbering is reliable.

### qwen_local backend — two-pass local LLM

Runs **Qwen3-4B-Instruct-2507** locally via Hugging Face `transformers` — no API, no token cost — (greedy decoding — more on why below) in two passes:

**① Main pass — chunked, multi-field**

_chunk_by_page → _build_prompt → _extract_chunk → _build_evidence → _resolve_field_candidates

Each page is split into groups of at most 5 elements ( `MAX_ELEMENTS_PER_CHUNK` ) — asking a 4B model to match ~15 candidates against 22 fields in one call measurably hurt recall. Each chunk's prompt lists every canonical field's own PLAN.md description (not just its name) and warns that a `table` -type element describes one repeated instance, never the whole document. Every citation is checked against its source element's actual text before counting as substantiated, and a citation from a `table` -type element is *always* unsubstantiated — a structural rule, not a prompt request, because the model can choose to ignore a prompt instruction but can't bypass a code-level check.

**② Second pass — per field, whole page**

_second_pass_missing_fields → _build_single_field_prompt → _extract_single_field

For whatever fields are still `missing` after the main pass, ask about exactly *one* field at a time against the *entire* page's elements (no further sub-chunking — a single-field prompt has no multi-field output budget to protect). This pass's prompt also *invites* a tentative citation from a `table` -type element instead of forbidding it outright, since the structural check above already guarantees it can never outrank a verified answer — it can only ever surface as `needs_review` .

Both passes resolve candidates through the same merge rule: candidates are only ever compared for conflict *within* a confidence tier. A verified citation always wins over a disagreeing unverifiable one; a low-confidence table guess can never drag a correctly-substantiated answer down to `conflicting` .

## Confidence score

Every field carries a `confidence` float — a triage signal for a reviewer, not a literal probability. It's a Pydantic `computed_field` on `FieldValue` , derived purely from `status` and `evidence` , so it can never drift out of sync with either one.

| STATUS | BAND | MOVES TOWARD CEILING WHEN… |
|---|---|---|
| conflicting | 0.15 – 0.29 | *fewer* disagreeing sources (more sources disagreeing is a worse sign) |

| STATUS | BAND | MOVES TOWARD CEILING WHEN… |
|---|---|---|
| ambiguous | 0.30 – 0.44 | more independent evidence flagging the same field as present-but-unclear |
| needs_review | 0.45 – 0.70 | more independent citations agreeing on the same unverified value |
| extracted | 0.85 – 1.00 | more independent, verified citations agreeing on the same value |
| missing | 0.00 | — no value, nothing to be confident about |

The bands never touch, so a `needs_review` field — no matter how much corroborating evidence it has — can never out-rank even a weakly-supported `extracted` one.

```
floor, ceiling = band for this status

if status == "conflicting":
    # more disagreement = worse, so corroboration runs backward
    corroboration = max(SATURATION_CONFUSED - evidence_count, 0) /
(SATURATION_CONFUSED - 2)
else:
    corroboration = max(0, min(evidence_count - 1, SATURATION - 1) / (SATURATION
- 1))

confidence = floor + (ceiling - floor) * corroboration
```

Worked example: `human_oversight` extracted from **3** independently-agreeing, verified citations lands at the top of the `extracted` band — **1.00**. The same field with just one citation would sit at the band's floor, **0.85**.

## Class & module reference

| MODULE | NAME | WHAT IT'S FOR |
|---|---|---|
| config | Settings | Model names, cache dir, vision on/off — read from `AI_OVERSIGHT_*` env vars via `Settings.from_env()`. Swapping models never touches code. |
| parsers.document_ir | Element | One parsed unit (text/key_value/table/chart) with its page, bbox, and content. The common currency every stage passes along. |
| parsers.document_ir | ParsedDocument | A document's full `Element` list plus parser warnings. Stage 1's output; Stage 2's input and output; Stage 3's input. |
| parsers.pdf_parser / pptx_parser | parse() | File → `ParsedDocument`. No model. Column-aware text, label/value pairing, chart-region and picture-shape detection. |

| MODULE | NAME | WHAT IT'S FOR |
|---|---|---|
| extractors.granite_vision | VisionObservation | One Granite call's raw response, parsed result, and status ( ok / needs_review ). |
| extractors.granite_vision | run_vision_pass() / enrich_document() | Finds the ambiguous regions, calls Granite, repairs truncated JSON, merges recovered content back into the IR as ordinary elements. |
| extractors.granite_vision | save_observations() / load_observations() | The caching pair: run the slow vision model once, replay its saved output for as many Stage-3 iterations as needed. |
| extractors.rules | extract() | Deterministic Stage-3 backend: field_id/label matching against the canonical 22, with a synonym table. |
| extractors.qwen_local | extract() | LLM Stage-3 backend: runs the main chunked pass, then the per-field second pass, then merges both into the result. |
| processing.merge | collect_metrics() / detect_performance_conflicts() | Backend-agnostic: harvests numeric metrics from chart/table elements and cross-checks a narrative claim against a chart's own numbers. |
| processing.validate | validate_field_shape() | Downgrades a value to needs_review if its shape doesn't match the field's expectation (e.g. risk_rating needs {inherent, residual} ). |
| schemas | FieldValue | One field's value, status, evidence list, and computed confidence . |
| schemas | Evidence | One citation: page/slide, element ID, quote, and the method that produced it. |
| schemas | ExtractionResult | The full output: 22 fields, metrics, incidents, review flags, warnings, and the raw vision observations. |
| schemas | ReviewFlag | A rule-generated suggestion for a human — explicitly never a compliance finding. |
| reporting.json_writer / markdown_writer | write() | Renders an ExtractionResult into the two sibling output files. |
| cli | parse / vision / extract | The three entry points: structural-only, vision-only (with caching), and the full pipeline. |

# CLI commands

**`ai_oversight parse <file> --out ir.json`**

Stage 1 only. No model. Its output JSON is reusable input to any Stage 3 backend later, via that backend's own loader.

**`ai_oversight vision <file> --out cache.json`**

Stage 2 only, saved to disk. Run once; reuse below as many times as needed.

**`ai_oversight extract <file> --backend qwen_local --vision-cache cache.json --out result.json`**

The full pipeline, Stage 3 reading from a saved vision cache instead of re-running Granite. Writes `result.json` and `result.review.md`. Drop `--vision-cache` and add `--vision` to run Granite live instead; drop both to skip Stage 2 entirely.

**`ai_oversight batch <file1> <file2> ... --backend qwen_local --out report.html`**

The same pipeline run independently over several documents, combined into one report: one row per (file, field), with that field's status, value, and confidence. Written as CSV, or as a styled HTML report when `--out` ends in `.html` — no extra flag needed. A file that fails to parse or extract is reported and skipped rather than aborting the whole batch.

## Testing with Ground-truth Check

Every file in this section is a **simulated** PDF or PPTX report — synthetic, fictional test documents built to exercise this pipeline, not real filings from any organization. Each one is run through the real pipeline via `batch` (live Granite Vision + `qwen_local`), and every resulting row is then checked by hand against that document's actual content: does this field's information really appear in the source, or not?

Two deliberately minimal files start this off — `Mini_Vision_Test_Internal_AI_Risk.pdf` (1 scanned page, 10 label/value facts) and `Mini_Vision_Test_External_AI_Usage.pptx` (1 slide, 9 facts) — producing 44 rows (22 fields × 2 files).

**37 of 44** rows are clean matches — the pipeline's `missing` correctly means "not stated anywhere," and its `extracted` correctly traces to a real, unambiguous statement. **6** are *Partial* cases: the source states something related but incomplete (e.g. only a residual risk rating, no inherent one) or one statement reasonably covers two related canonical fields (a single generic "Owner" line for both `business_owner` and `technical_owner`). **1** is the interesting case — `next_review_date` on the PDF — where the model inferred a plausible-looking date from the validation statement's own date, a genuinely different fact; the substantiation check caught the mismatch and correctly held it at `needs_review` instead of letting it through as `extracted`.

### Mini_Vision_Test_Internal_AI_Risk.pdf (1 page)

Granite Vision this file has no native text layer (fully scanned) -- every fact below, including every **Source** quote, was recovered via Granite Vision's `tables_json` task, not parsed natively.

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| submission_id | missing | | 0.0 | Not stated | |
| reporting_entity | missing | | 0.0 | Not stated | |
| application_name | missing | | 0.0 | Not stated | |
| business_use_case | extracted | Settlement exception prediction | 0.85 | Yes | Granite Vision Use case: Settlement exception prediction |
| business_owner | extracted | Post-Trade Operations | 0.85 | Partial | Granite Vision Owner: Post-Trade Operations (one generic “Owner” row, not split business/technical) |
| technical_owner | extracted | Post-Trade Operations | 0.85 | Partial | Granite Vision Owner: Post-Trade Operations (same row reused) |
| model_type | extracted | Gradient-boosted trees | 0.85 | Yes | Granite Vision Model type: Gradient-boosted trees |
| model_provider | extracted | Internal / in-house | 0.85 | Yes | Granite Vision Provider: Internal / in-house |
| model_version | missing | | 0.0 | Not stated | |
| lifecycle_status | missing | | 0.0 | Not stated | |
| first_deployment_date | missing | | 0.0 | Not stated | |
| input_data | extracted | Confidential transaction records | 0.85 | Partial | Granite Vision Data sensitivity: Confidential transaction records (no distinct data-sources statement) |
| data_sensitivity | extracted | Confidential transaction records | 0.85 | Yes | Granite Vision Data sensitivity: Confidential transaction records |
| financial_criticality | extracted | HIGH | 0.85 | Yes | Granite Vision Financial criticality: HIGH |
| human_oversight | extracted | Analyst approves any action | 0.85 | Yes | Granite Vision Human oversight: Analyst approves any action |
| validation_status | extracted | Passed independent testing, Oct 2026 | 0.85 | Yes | Granite Vision Validation results: Passed independent testing, Oct 2026 |
| model_performance | missing | | 0.0 | Not stated | |
| risk_rating | needs_review | MEDIUM | 0.45 | Partial | Granite Vision Residual risk rating: MEDIUM (no inherent rating stated) |
| incident_history | extracted | 1 low-severity alert issue, Sep 2026 | 0.85 | Yes | Granite Vision Incident history: 1 low-severity alert issue, Sep 2026 |

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| monitoring_frequency | missing | | 0.0 | Not stated | |
| fallback_process | missing | | 0.0 | Not stated | |
| next_review_date | needs_review | 2026-10-01 | 0.45 | Not stated | Granite Vision model inferred this from the Oct 2026 validation date -- a different fact; correctly caught as needs_review, not extracted |

## Mini_Vision_Test_External_AI_Usage.pptx (1 slide)

Granite Vision this slide is a single picture shape -- every fact below, including every **Source** quote, was recovered via Granite Vision's `tables_json` task, not parsed natively.

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| submission_id | missing | | 0.0 | Not stated | |
| reporting_entity | extracted | HarborView Operations (fictional) | 0.85 | Yes | Granite Vision Reporting organization: HarborView Operations (fictional) |
| application_name | missing | | 0.0 | Not stated | |
| business_use_case | missing | | 0.0 | Not stated | |
| business_owner | missing | | 0.0 | Not stated | |
| technical_owner | missing | | 0.0 | Not stated | |
| model_type | extracted | Hosted instruction LLM | 0.85 | Yes | Granite Vision Model / provider: Hosted instruction LLM / Northstar AI |
| model_provider | extracted | Northstar AI | 0.85 | Yes | Granite Vision Model / provider: Hosted instruction LLM / Northstar AI |
| model_version | missing | | 0.0 | Not stated | |
| lifecycle_status | missing | | 0.0 | Not stated | |
| first_deployment_date | missing | | 0.0 | Not stated | |
| input_data | extracted | Restricted, masked customer fields | 0.85 | Partial | Granite Vision Data classification: Restricted, masked customer fields (no distinct data-sources statement) |
| data_sensitivity | extracted | Restricted, masked customer fields | 0.85 | Yes | Granite Vision Data classification: Restricted, masked customer fields |
| financial_criticality | extracted | MEDIUM | 0.85 | Yes | Granite Vision Financial criticality: MEDIUM |

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| human_oversight | extracted | Mandatory approval before any action | 0.85 | Yes | Granite Vision Human oversight: Mandatory approval before any action |
| validation_status | extracted | Passed review, October 2026 | 0.85 | Yes | Granite Vision Validation results: Passed review, October 2026 |
| model_performance | missing | | 0.0 | Not stated | |
| risk_rating | needs_review | MEDIUM | 0.45 | Partial | Granite Vision Residual risk rating: MEDIUM (no inherent rating stated) |
| incident_history | extracted | 0 reportable in past 90 days | 0.85 | Yes | Granite Vision Incidents: 0 reportable in past 90 days |
| monitoring_frequency | extracted | Monthly | 0.85 | Yes | Granite Vision Monitoring frequency: Monthly |
| fallback_process | missing | | 0.0 | Not stated | |
| next_review_date | missing | | 0.0 | Not stated | |

## Sample_Internal_AI_Oversight_Report.pdf (4 pages)

Every one of the 22 fields genuinely exists here, cleanly labeled in an explicit field-ID inventory on page 2 ("22 mapped fields • explicit IDs support extraction testing") -- so **Ground truth** is Yes throughout, and after fixing two real precision bugs found via this exact fixture (preferring a structured element's own label over a bare fallback value, and preferring an explicit key_value citation over a disagreeing narrative mention elsewhere in the document), the pipeline now resolves **all 22 of 22** fields correctly.

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| submission_id | extracted | AI-INT-2026-041 | 0.93 | Yes | Submission ID: AI-INT-2026-041 |
| reporting_entity | extracted | Example Post-Trade Operations (fictional) | 0.85 | Yes | Legal entity: Example Post-Trade Operations (fictional) |
| application_name | extracted | Settlement Failure Early Warning (SFEW) | 0.93 | Yes | Application name: Settlement Failure Early Warning (SFEW) |
| business_use_case | extracted | T+1 settlement exception prediction | 0.93 | Yes | Business use case: T+1 settlement exception prediction |
| business_owner | extracted | Maya Chen, Director, Post-Trade Analytics | 0.85 | Yes | Business owner: Maya Chen, Director, Post-Trade Analytics |
| technical_owner | extracted | Alex Rivera, Senior ML Engineer | 0.85 | Yes | Technical owner: Alex Rivera, Senior ML Engineer |

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| model_type | extracted | Gradient-boosted trees (XGBoost) | 0.85 | Yes | Model type: Gradient-boosted trees (XGBoost) |
| model_provider | extracted | Internally developed; no external inference API | 0.85 | Yes | AI provider: Internally developed; no external inference API |
| model_version | extracted | 2.3.1 | 0.93 | Yes | Model version: 2.3.1 |
| lifecycle_status | extracted | Production | 0.93 | Yes | Lifecycle status: Production |
| first_deployment_date | extracted | 2026-07-15 | 0.85 | Yes | First deployment: 2026-07-15 |
| input_data | extracted | Transaction metadata + exception history | 0.85 | Yes | Input data: Transaction metadata + exception history |
| data_sensitivity | extracted | Confidential, transaction-level data | 0.93 | Yes | Data sensitivity: Confidential, transaction-level data |
| financial_criticality | extracted | High; advisory operations prioritization | 0.85 | Yes | Financial criticality: High; advisory operations prioritization |
| human_oversight | extracted | Analyst approves every escalation | 1.0 | Yes | Human oversight: Analyst approves every escalation |
| validation_status | extracted | Passed independent validation on 2026-10-20 | 0.93 | Yes | Validation status: Passed independent validation on 2026-10-20 |
| model_performance | extracted | {'precision': 0.91, 'recall': 0.86, 'roc_auc': 0.95} | 1.0 | Yes | Performance: Precision 0.91; recall 0.86; ROC-AUC 0.95 |
| risk_rating | extracted | {'inherent': 'HIGH', 'residual': 'MEDIUM'} | 1.0 | Yes | Risk rating: Inherent HIGH; residual MEDIUM |
| incident_history | extracted | 1 closed incident; see page 4 | 1.0 | Yes | Incident history: 1 closed incident; see page 4 |
| monitoring_frequency | extracted | Daily metrics; monthly drift review | 1.0 | Yes | Monitoring frequency: Daily metrics; monthly drift review |
| fallback_process | extracted | Manual rules-based exception queue | 0.93 | Yes | Fallback process: Manual rules-based exception queue |
| next_review_date | extracted | 2027-01-20 | 0.85 | Yes | Next periodic review: 2027-01-20 |

## Sample_External_AI_Usage_Disclosure.pptx (7 slides)

Same pattern as the PDF above -- an explicit field-ID inventory across slides 3-4 means every field is genuinely present ( Yes throughout), and after the same two fixes, the pipeline resolves **all 22 of 22** fields correctly

here too.

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| submission_id | extracted | EXT-2026-118 | 0.93 | Yes | Submission ID: EXT-2026-118 |
| reporting_entity | extracted | Meridian Clearing Services (fictional) | 0.93 | Yes | Reporting organization: Meridian Clearing Services (fictional) |
| application_name | extracted | Collateral Exception Triage Copilot (CETC) | 0.85 | Yes | AI application: Collateral Exception Triage Copilot (CETC) |
| business_use_case | extracted | Collateral exception classification + routing | 1.0 | Yes | Business use case: Collateral exception classification + routing |
| business_owner | extracted | Priya Nair, Director of Operations Risk | 0.85 | Yes | Business owner: Priya Nair, Director of Operations Risk |
| technical_owner | extracted | Ethan Brooks, ML Platform Lead | 0.85 | Yes | Technical owner: Ethan Brooks, ML Platform Lead |
| model_type | extracted | Retrieval-augmented LLM + classifier | 0.93 | Yes | Model type: Retrieval-augmented LLM + classifier |
| model_provider | extracted | NimbusAI Cloud (fictional hosted provider) | 0.93 | Yes | AI / model provider: NimbusAI Cloud (fictional hosted provider) |
| model_version | extracted | CETC v1.7; NimbusText-4.2 | 0.85 | Yes | Version: CETC v1.7; NimbusText-4.2 |
| lifecycle_status | extracted | Controlled rollout / analyst-in-the-loop | 0.85 | Yes | Deployment status: Controlled rollout / analyst-in-the-loop |
| first_deployment_date | extracted | 2026-08-04 | 0.85 | Yes | First deployed: 2026-08-04 |
| input_data | extracted | Tokenized tickets, settlement messages, runbooks | 0.93 | Yes | Data inputs: Tokenized tickets, settlement messages, runbooks |
| data_sensitivity | extracted | Confidential; direct identifiers tokenized | 1.0 | Yes | Data sensitivity: Confidential; direct identifiers tokenized |
| financial_criticality | extracted | Moderate; human-controlled collateral operations | 0.85 | Yes | Financial criticality: Moderate; human-controlled collateral operations |
| human_oversight | extracted | Analyst must approve every classification & action | 1.0 | Yes | Human oversight: Analyst must approve every classification & action |
| validation_status | extracted | 2026-10-08; 600-case independent review | 1.0 | Yes | Validation: 2026-10-08; 600-case independent review |

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| model_performance | extracted | {'accuracy': 0.92, 'grounded_response_rate': 0.97} | 1.0 | Yes | Model performance: Accuracy 92%; grounded response rate 97% |
| risk_rating | extracted | {'inherent': 'HIGH', 'residual': 'MEDIUM'} | 1.0 | Yes | Risk rating: Inherent HIGH; residual MEDIUM |
| incident_history | extracted | 2 resolved incidents (Sep 07, Oct 02, 2026) | 1.0 | Yes | Incident history: 2 resolved incidents (Sep 07, Oct 02, 2026) |
| monitoring_frequency | extracted | Daily prompt logs; weekly QA sampling | 1.0 | Yes | Monitoring frequency: Daily prompt logs; weekly QA sampling |
| fallback_process | extracted | Manual triage queue + static decision tree | 1.0 | Yes | Fallback process: Manual triage queue + static decision tree |
| next_review_date | extracted | 2027-01-08 | 0.93 | Yes | Next review date: 2027-01-08 |

## Vision_Required_Internal_AI_Risk_Report.pdf (4 pages)

The original, full-size fixture that motivated Phase 3: every page is a scanned image with no native text layer, so every row below comes from Granite Vision. Most facts are genuinely present and cleanly stated (13 of 22 fields) -- these mostly resolve correctly. Three fields ( `risk_rating` , `monitoring_frequency` , `fallback_process` ) are only *Partial*: the document states something related but incomplete (e.g. a residual rating with no inherent one) or buried inside one row of a 5-row control matrix rather than stated as a document-level fact. The remaining 6 are genuinely absent. Two results are worth calling out specifically: `technical_owner` has no real answer in this document at all, and the structural table-citation check correctly stopped an irrelevant control-matrix citation from being confidently accepted; `monitoring_frequency` came back `missing` this run even though "Monthly PSI review" is a real (if buried) signal -- the per-field second pass has surfaced it in some runs and not others, a known soft spot for single-row facts hidden inside a multi-record table.

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| submission_id | missing | | 0.0 | Not stated | |
| reporting_entity | missing | | 0.0 | Not stated | |
| application_name | extracted | Settlement Failure Early Warning | 0.85 | Yes | Granite Vision APPLICATION: Settlement Failure Early Warning |
| business_use_case | extracted | Predict and prioritize likely US equity settlement failures | 0.85 | Yes | Granite Vision USE CASE: Predict and prioritize likely US equity settlement failures |
| business_owner | needs_review | Maya Chen, Director of Settlement Risk | 0.45 | Yes | Granite Vision BUSINESS OWNER: Maya Chen, Director of Settlement Risk |

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| technical_owner | needs_review | Operations Lead | 0.45 | Not stated | Granite Vision no distinct technical-owner row exists; the model's "Operations Lead" citation is a control's owner in an unrelated 5-row control matrix |
| model_type | extracted | LightGBM gradient boosting + rule guardrails | 0.85 | Yes | Granite Vision AI/MODEL TYPE: LightGBM gradient boosting + rule guardrails |
| model_provider | extracted | Internally developed / LightGBM v4.5.0 | 0.85 | Yes | Granite Vision MODEL PROVIDER: Internally developed / LightGBM v4.5.0 |
| model_version | extracted | r2.3.1 | 0.85 | Yes | Granite Vision MODEL VERSION: r2.3.1 |
| lifecycle_status | extracted | Production | 0.85 | Yes | Granite Vision DEPLOYMENT STATUS: Production |
| first_deployment_date | needs_review | 2026-04-15 | 0.45 | Yes | Granite Vision PRODUCTION DATE: 15 April 2026 |
| input_data | extracted | Synthetic settlement events; exception queues; counterparty flags | 0.85 | Yes | Granite Vision DATA SOURCES: Synthetic settlement events; exception queues; counterparty flags |
| data_sensitivity | extracted | Confidential market participant transaction data | 0.85 | Yes | Granite Vision DATA CLASSIFICATION: Confidential market participant transaction data |
| financial_criticality | extracted | HIGH - impacts exception prioritization | 0.85 | Yes | Granite Vision FINANCIAL CRITICALITY: HIGH - impacts exception prioritization |
| human_oversight | extracted | Operations analyst signs off on interventions | 0.85 | Yes | Granite Vision HUMAN OVERSIGHT: Operations analyst signs off on interventions |
| validation_status | missing | | 0.0 | Not stated | |
| model_performance | extracted | {'precision': 0.9109999999999999, 'recall': 0.87, 'samples': 1000.0} | 0.85 | Yes | Granite Vision n=1,000 records \| Precision 91.1% \| Recall 87.0% |
| risk_rating | needs_review | MEDIUM | 0.45 | Partial | Granite Vision RESIDUAL RISK: MEDIUM (no inherent rating stated) |
| incident_history | missing | | 0.0 | Not stated | |
| monitoring_frequency | missing | | 0.0 | Partial | Granite Vision C-03 \| Monthly PSI review (one control among 5, not a document-level statement) |
| fallback_process | needs_review | Manual fallback drill | 0.45 | Partial | Granite Vision C-04 \| Manual fallback drill (one control among 5, not a document-level statement) |
| next_review_date | missing | | 0.0 | Not stated | |

## Vision_Required_External_AI_Usage_Disclosure.pptx (6 slides)

Same vision-only character as the PDF above. All **12** fields the pipeline marks `extracted` are genuine, correct matches, and **4 of 5** genuinely absent fields are correctly `missing` -- the fifth, `technical_owner`, is again a safety-net catch (an irrelevant control-matrix citation correctly held at `needs_review` instead of accepted). After the key_value-citation fix, `risk_rating` no longer comes back `conflicting`: the model now synthesizes both the inherent and residual ratings into one combined answer, correctly held at `needs_review` rather than `extracted` since its own citation only points to one of the two source rows. One open limitation remains: `incident_history` comes back `ambiguous` rather than a confident count, since the model can see a real 3-row incident table but doesn't confidently aggregate it into one document-level answer.

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| submission_id | extracted | EXT-MRG-2026-031 | 0.85 | Yes | Granite Vision APPLICATION ID: EXT-MRG-2026-031 |
| reporting_entity | extracted | Riverton Clearing Services (fictional) | 0.85 | Yes | Granite Vision SUBMITTING FIRM: Riverton Clearing Services (fictional) |
| application_name | extracted | Margin Call Support Copilot | 0.85 | Yes | Granite Vision APPLICATION NAME: Margin Call Support Copilot |
| business_use_case | extracted | Draft margin-call explanations and suggest resolution codes | 0.93 | Yes | Granite Vision BUSINESS USE CASE: Draft margin-call explanations and suggest resolution codes |
| business_owner | extracted | Elena Ortiz, VP Collateral Services | 0.85 | Yes | Granite Vision BUSINESS OWNER: Elena Ortiz, VP Collateral Services |
| technical_owner | needs_review | AI Governance | 0.45 | Not stated | Granite Vision no distinct technical-owner row exists; the model's "AI Governance" citation is a control's owner in an unrelated 5-row control matrix |
| model_type | extracted | Retrieval-augmented LLM, instruction tuned | 0.85 | Yes | Granite Vision MODEL TYPE: Retrieval-augmented LLM, instruction tuned |
| model_provider | extracted | Self-hosted Qwen2.5-7B-Instruct | 0.85 | Yes | Granite Vision MODEL PROVIDER: Self-hosted Qwen2.5-7B-Instruct |
| model_version | extracted | mc-support-v1.6.3 | 0.85 | Yes | Granite Vision MODEL VERSION: mc-support-v1.6.3 |
| lifecycle_status | needs_review | PILOT | 0.45 | Partial | Granite Vision "PILOT" appears in a garbled, misaligned vision-recovered table, not a clean document-level statement |
| first_deployment_date | missing | | 0.0 | Not stated | |
| input_data | extracted | Approved margin statements, collateral rules, dispute history | 0.85 | Yes | Granite Vision DATA SOURCES: Approved margin statements, collateral rules, dispute history |
| data_sensitivity | extracted | RESTRICTED - client margin and account-level data | 0.85 | Yes | Granite Vision DATA SENSITIVITY: RESTRICTED - client margin and account-level data |

| FIELD | STATUS | VALUE | CONF. | GROUND TRUTH | SOURCE |
|---|---|---|---|---|---|
| financial_criticality | extracted | HIGH - supports same-day margin exception workflows | 0.93 | Yes | Granite Vision FINANCIAL CRITICALITY: HIGH - supports same-day margin exception workflows |
| human_oversight | extracted | Mandatory analyst approval for all client messages | 1.0 | Yes | Granite Vision HUMAN OVERSIGHT: Mandatory analyst approval for all client messages |
| validation_status | missing | | 0.0 | Not stated | |
| model_performance | needs_review | {'triage_accuracy': '86%', 'grounded_response_rate': '92%', 'accuracy_target': '90%'} | 0.45 | Yes | Granite Vision Triage accuracy/Grounded response rate chart: 86%/92%, 89%/95% vs. 90% target |
| risk_rating | needs_review | INHERENT RISK: HIGH, RESIDUAL RISK: MEDIUM | 0.7 | Yes | Granite Vision INHERENT RISK: HIGH / RESIDUAL RISK: MEDIUM (two separate rows, both stated) |
| incident_history | ambiguous | | 0.3 | Yes | Granite Vision 3-row incident table: INC-103, INC-102, INC-101 |
| monitoring_frequency | needs_review | Daily quality check / monthly review | 0.45 | Partial | Granite Vision Model drift \| Daily quality check / monthly review (one control among 5, not a document-level statement) |
| fallback_process | missing | | 0.0 | Not stated | |
| next_review_date | missing | | 0.0 | Not stated | |

ai_oversight — local AI-governance-report extraction. Every `extracted` value traces to real source text; everything else is marked for a human to look at.